

Cryptocurrency Market Analysis

This project focuses on the analysis of the dynamics of various popular cryptocurrencies in the market. With examining the price trends, volatility, and relationship between Bitcoin, Ethereum, XRP, and Dogecoin. The goal of this analysis is to understand better the behavior of these cryptocurrencies over time and how they compare over time. Crypto is known for yielding high returns as well as rough drops for individuals, which makes for interesting visuals and analysis, especially as they continue to increase in popularity and understanding. To support the analysis, a Streamlit app was created to provide a visually appealing and interactive way to view the results found, this app allows the user to easily explore the data.

The analysis showed key patterns across the chosen cryptocurrencies. Bitcoin for the most part showed consistency in day to day movement but had occasionally large spikes. With the overall consistent movement, it would yield a more narrow range of return if it wasn't sold in the correct timing of one of the spikes. Ethereum in comparison, had a larger variation in its day to day returns. Which displays more common fluctuation, but also has less very extreme outliers/spikes, especially when compared to XRP and Dogecoin. XRP and Dogecoin displayed similar behaviors, with both of them having more moderate day to day variation and more frequent spikes in returns. Both at least tripled their returns at some point in time when compared to Bitcoin and Ethereum. In addition, the correlation analysis showed that most of the cryptocurrencies tend to move in similar directions over time, especially Bitcoin and Ethereum, suggesting that they are influenced by similar market factors. The use of a logarithmic scale also highlighted how the smaller-value cryptocurrencies like XRP and Dogecoin still follow similar growth trends to Bitcoin and Ethereum, even though their prices differ significantly.

There are multiple different limitations to the analysis. One of the biggest limitations is the data only going to 2021. With the growth and popularity growing in the technological field overall, and the increase in how people consume information via the internet and social media has only increased since 2021. Additionally, only 4 different cryptocurrencies were analyzed which is not close to fully representing the full broader market of cryptocurrencies as a whole. Limitations in modeling as historical data is being used rather than the use of a predictive model, also the dataset does not include any external factors that could influence prices, such as general economic conditions, government regulations, the Covid-19 pandemic, etc. Being able to visually see where the cryptocurrencies are predicted to head towards would be a great benefit to anticipate future outcomes and results.